Question	Answer	Mark	Mark scheme	Additional guidance
13	$3 : \sqrt{30}$	P1	for start of process to find a ratio to aid comparison, eg ratio of lengths **A** : **B**, eg $\sqrt[3]{27} : \sqrt[3]{64}$ (= 3 : 4) or 3, 4 oe or ratio of lengths **B** : **C**, eg $\sqrt{8} : \sqrt{15}$ or $\sqrt{8}$, $\sqrt{15}$ oe	
		P1	for process to find 2 ratios which can be combined eg, ratio of lengths **A** : **B**, eg $\sqrt[3]{27} : \sqrt[3]{64}$ (= 3 : 4) or 3, 4 oe and ratio of lengths **B** : **C**, eg $\sqrt{8} : \sqrt{15}$ or $\sqrt{8}$, $\sqrt{15}$ **or** for process to find ratio of surface areas of **A** and **B**, eg 9 : 16 oe	
		P1	for process to use multiplier to combine ratios of lengths, eg ratio of lengths **A** : **B** multiplied by: $\frac{1}{\sqrt{2}}$, $3 \times \frac{1}{\sqrt{2}} : 4 \times \frac{1}{\sqrt{2}}$) oe or ratio of lengths **B** : **C** multiplied by: $\sqrt{2}$, $\sqrt{8} \times \sqrt{2} : \sqrt{15} \times \sqrt{2}$ oe **or** for process to use multiplier to combine ratios of surface areas, eg ratio of areas **A** : **B** multiplied by: $\frac{1}{2}$, $9 \times \frac{1}{2} : 16 \times \frac{1}{2}$ (= 4.5 : 8) oe or ratio of areas **B** : **C** multiplied by: 2, $8 \times 2 : 15 \times 2$ (= 16 : 30) oe	
		A1	for $3 : \sqrt{30}$. oe	